Reg. No: …….

Name: …….

MAHATHMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2025

THIRD SEMESTER

Discipline Specific Core Course (DSE) MG3DSEFSC200-INSTRUMENTATION-CHEMICAL

FORENSIC SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours
Maximum Mark:70

Remember (K), Understanding(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skills(S), Interest(I) and Appreciation (AP)

Students should attempt at least one question from each course outcome to enhance their overall outcomes attainability

Part A

Short Answer Type Questions
Answer all**10** questions
Each question carries 1 mark
One Word Answer Questions (1 Mark)

1. What is electromagnetic radiation?*[U] / [CO4]*
2. What is fluorescence? *[U] / [CO4]*
3. State one difference between absorption and emission.*[U] / [CO2]*
4. What is the unit of chemical shift?*[U] / [CO3]*
5. What is the full form of AAS?*[U] / [CO2]*
6. Which detector converts ions into an electric signal in MS?*[U] / [CO3]*
7. What is the ratio used in calculating Rf value?*[U] / [CO2]*
8. Is HPTLC a qualitative or quantitative technique?*[U] / [CO2]*
9. What type of scan mode collects selected ion data?*[U] / [CO5]*
10. What type of elution shows the varying solvent composition?*[U] / [CO3]*

Answer any **5** questions

Short answer questions (3Marks)

Each question carries 3 marks

1. Explain the interaction between matter and radiation.*[U] / [CO1]*
2. Define and differentiate between reflection and refraction.*[U] / [CO1]*
3. What is the basic principle of AAS?*[U] / [CO1]*
4. Mention any three forensic applications of HPLC.*[U] / [CO3]*
5. What is the spin-spin coupling in NMR Spectroscopy?*[U] / [CO3]*
6. What is the principle of paper chromatography?*[U] / [CO4]*
7. Define Rf value. Also state its use in TLC?*[U] / [CO5]*

Short Essay (6 Marks)

Answer any **5** questions

1. Describe the processes of reflection, scattering and transmission.*[U] / [CO1]*
2. Describe the concepts of absorption and emission with examples.*[U] / [CO2]*
3. Describe the principle and forensic use of GC-MS.*[U] / [CO4]*
4. Write a note on types of detectors used in HPLC.*[U] / [CO3]*
5. Explain instrumentation of Atomic Absorption Spectroscopy.*[U] / [CO3]*
6. Define buffers and explain their importance in maintaining pH in biological systems.*[U] / [CO4]*
7. Explain the theory behind separation techniques used in chromatography.*[U] / [CO5]*

Essay (15 Marks)

Answer any **1** question

1. Elaborate on the interaction between electromagnetic radiation and matter. Include all major processes and classification of radiation.*[A] / [CO3]*

2. Discuss the principle, instrumentation, scanning modes, and forensic applications of Mass Spectrometry.*[A] / [CO3]*